

Hokyun Lee

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EDUCATION

Seoul National University Ph.D. candidate in Department of Intelligence and Information, Advisor: Jaeheung Park	Seoul, South Korea 2023–Current
Seoul National University A combined M.S./Ph.D. program in Department of Intelligence and Information. – GPA: 3.8/4.3	Seoul, South Korea 2021–Current
Daegu Gyeongbuk Institute of Science and Technology (DGIST) B.E. in the School of Undergraduate Studies – GPA: 3.7/4.3	Dageu, South Korea 2017–2021

EXPERIENCE

Seoul National University/DYROS Lab. System Engineer/Dynamic Robotic Systems Laboratory – ANA AVATAR XPRIZE COMPETITION – Developing the TOCABI avatar system integrating control, haptics, sensing, and design.	Seoul, South Korea 2020–2022
Daegu Gyeongbuk Institute of Science and Technology Control Engineer/Bio Robotics and Mechatronics Laboratory, Advisor: Dongwon Yun – Development of a Quadruped Walking Robot for Campus Safety at DGIST – A quadruped robot for campus safety at DGIST through autonomous patrol, hazard detection.	Dageu, South Korea 2020
Daegu Gyeongbuk Institute of Science and Technology System Engineer/Motion Control Laboratory, Advisor: Sehoon Oh – Development of a Robotic Leg for High-Speed Quadruped Robot – Minimal-Impact Locomotion of a Robotic Leg Using a Modular Series Elastic Actuator	Dageu, South Korea 2019

PUBLICATIONS

- [1] **H. Lee**, W.-J. Baek, J. Cha, and J. Park, “Tolebi: Learning fault-tolerant bipedal locomotion via online status estimation and fallibility rewards”, *arXiv preprint arXiv:2602.05596*, 2026 (Accepted for Publication at IEEE International Conference on Robotics and Automation (ICRA) 2026).
- [2] J. Ahn, J. Jung, Y. Lee, **H. Lee**, S. Haddadin, and J. Park, “Efficient computation of whole-body control utilizing simplified whole-body dynamics via centroidal dynamics”, *International Journal of Control, Automation and Systems*, vol. 23, no. 9, pp. 2656–2665, 2025.
- [3] D. Kim, **H. Lee**, J. Cha, and J. Park, “Bridging the reality gap: Analyzing sim-to-real transfer techniques for reinforcement learning in humanoid bipedal locomotion”, *IEEE Robotics & Automation Magazine*, 2025.
- [4] J. Kim, **H. Lee**, and J. Park, “Real-time whole-body model predictive control for bipedal locomotion with a novel kino-dynamic model and warm-start method”, *arXiv preprint arXiv:2505.19540*, 2025.

- [5] B. Park, D. Kim, D. Lim, S. Park, J. Ahn, S. Kim, J. Shin, E. Sung, J. Sim, J. Kim, M.-J. Kim, J. Cha, G. Park, **H. Lee**, *et al.*, “Intuitive and interactive robotic avatar system for tele-existence: Team snu in the ana avatar xprize finals”, *International Journal of Social Robotics*, vol. 17, no. 3, pp. 379–407, 2025.
- [6] J. Ahn, J. Jung, **H. Lee**, and J. Park, “Time-efficient contact consistent whole-body control framework via reduced-dimension dynamics construction”, *IEEE Access*, vol. 13, pp. 2286–2297, 2024.
- [7] **H. Lee**, G. Park, J. Shin, B. Park, and J. Park, “Foot-operated telelocomotion interface for avatar robots utilizing mecanum wheel-based mobile platforms”, in *2023 23rd International Conference on Control, Automation and Systems (ICCAS)*, IEEE, 2023, pp. 651–656.
- [8] B. Park, J. Jung, J. Sim, S. Kim, J. Ahn, D. Lim, D. Kim, M. Kim, S. Park, E. Sung, **H. Lee**, *et al.*, “Team snu’s avatar system for teleoperation using humanoid robot: Ana avatar xprize competition”, in *RSS 2022 workshop on “towards robot avatars: perspectives on the ANA avatar XPRIZE competition*, 2022.
- [9] J. Kim, B. Park, **H. Lee**, and J. Park, “Hybrid position/torque ankle controller for minimizing zmp error of humanoid robot”, in *2021 18th International Conference on Ubiquitous Robots (UR)*, IEEE, 2021, pp. 211–216.

TEACHING

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| <ul style="list-style-type: none"> • Head Teaching Assistant at Seoul National University
<i>DYROS Robotics Boot Camp II (Robot Design, SI, Simulation and Control)</i> • Teaching Assistant at Seoul National University
<i>DYROS Robotics Boot Camp II (Robot Design, SI, Simulation and Control)</i> • Teaching Assistant at Seoul National University
<i>DYROS Robotics Boot Camp I (ROS: Robot Operating System)</i> | <p>2023–2026</p> <p>2022–2023</p> <p>2022–2026</p> |
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SKILLS

- **Software:** C++, Python, MATLAB, PyTorch, Numpy, Anaconda, Docker, OpenCV, C
- **Hardware:** Fusion360, 3D Printing, Wiring, Soldering
- **System Integration:** RTLinux, ROS1, ROS2, EtherCAT, Shared Memory, Multi-threading, Qt, RPi, Jetson series
- **Applications:** Isaac Gym/Lab, MuJoCo, CoppeliaSim, MoveIt, Git, VS code, Notion, Slack and Agents..

LANGUAGES

- **Language:** Korean (mother tongue)
- **Language:** English (intermediate)